

1. Basic Principles of Probability Theory and Statistical Inference

- 1.1 Basic Concepts in Probability & Statistics
- 1.2 Random Variable
- 1.3 Probability Distribution
- 1.4 Statistical Estimator
- 1.5 Hypothesis Testing

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1. Basic Principles of Probability Theory and Statistical Inferences

1.1. Basic Concepts in Probability & Statistics

Statistics (统计学): a science of data analysis.

Study on scientific methods for sampling, collection, organization, summarization, presentation and analysis of data.

Study on the scientific methods to make effective inferences and reasonable conclusions on population according to the analysis results of data.

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Population (总体): The collection of all individuals for a surveyed object

In real statistical analysis, it is usually impossible to survey all individuals, but rather some individuals sampled from population.

Sample (样本): the collection of individuals analyzed in real study

We can make inferences on the properties of population via analysis of sample data

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The Essential of Statistics

Describe the feature of population

参数(parameter)

样本(sample)

估计方法: 点估计 (point estimate)

离散变量(discrete variable);
连续变量(continuous variable);

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群体 样本

Population parameter & statistic

population (limited, unlimited) → sampling → Sample (limited) (样本2, 样本3)

Parameter which is used to describe properties of population:

Population mean: $\mu = \lim_{n \rightarrow \infty} (\frac{1}{n} \sum_{i=1}^n X_i)$

Population variance: $\sigma^2 = \lim_{n \rightarrow \infty} (\frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2)$

Statistic: use sample data to estimate population parameter

estimate $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$

The value of statistic will be different for different samples, thus, there is variation (variance) for a statistic across samples

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均值, 方差

Experiment (试验): The process of investigating and collecting data according to the prior designed procedure

Any possible outcome from performance of experiment is called a Outcome (结局) of experiment. If same experiment is performed repeatedly, different outcomes will be produced.

Event (事件): One or some outcomes of experiment

Event usually denoted by capital letter A, B, C.

Sample space (样本空间): All possible outcomes of experiment

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1.2. Random Variable

The purpose of statistical analysis is to infer property of population

Parameter (参数): a quantity describing characteristics of population. They are usually unknown, but can be inferred via statistical analysis.

Statistic (统计量): a quantity describing characteristics of a sample

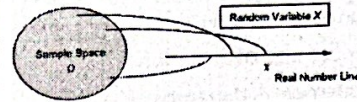
Random variable (随机变量): A random variable is a variable whose domain is the set of basic events, and whose range (outcome) could be numerical or categorical.

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Random variable(随机变量):

A random variable is a real-valued function of the experimental outcome



Random variable usually is denoted by capital letter, for example, X, Y, Z . Their real result or numerical values are denoted by lowercase letter, for example, x, y, z

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Discrete variable(离散变量): random variable expressed in count number

Continuous variable(连续变量): random variable expressed in any real number

Between random variables often have different degrees of relationship, which could be represented by mathematical model or mathematical function.

Linear model(线性模型): mathematical function that describes relationship between variables, its parameters have only a simple linear relationship. The theory of linear model is the basis of most widely used statistical analysis methods, such as regression analysis, correlation analysis, variance analysis and covariance analysis.

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Probability density/mass function(概率密度函数, pdf / pmf): the probability that random variable X takes value of x in repeated experiments,

$$f(x) = P(X = x)$$

The probability density function has the following properties

(1) for all x , $0 \leq f(x) \leq 1$

(2) For discrete variable $\sum f(x) = 1$

(3) for continuous variable $\int f(x) dx = 1$

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Distribution function(分布函数): the accumulated probability when random variable X is less than x in repeated experiments,

$$F(x) = P(X \leq x)$$

The distribution function has the following properties

(1) $\lim_{x \rightarrow -\infty} F(x) = 0$ (4) If $a < b$, then $F(a) \leq F(b)$

(2) $\lim_{x \rightarrow \infty} F(x) = 1$ (5) $P(a < X \leq b) = F(b) - F(a)$

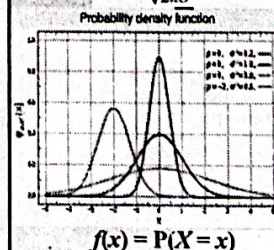
(3) $\lim_{h \rightarrow 0} F(x+h) = F(x)$

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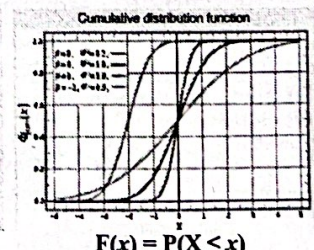
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Normal distribution

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < +\infty \quad (1.8)$$



$$f(x) = P(X=x)$$



$$F(x) = P(X \leq x)$$

$$= \int_{-\infty}^x f(x) dx$$

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The expected value (期望值) of a random variable :

In an experiment, if a random variable X has probability density function $f(x)$, then the expectation of X is defined by

$$E(X) = \sum_x x f(x) \text{ if } X \text{ is a discrete random variable}$$

$$E(X) = \int_{-\infty}^{+\infty} x f(x) dx \text{ if } X \text{ is a continuous random variable}$$

The property of the 1-order expectation (the center of the distribution)

? $E(g(X))$, such as $E(X^2)$, $E(3X^3 - 2X^2 + 1)$

$$E(g(X)) = \int_{-\infty}^{+\infty} g(x) f(x) dx$$

According to the definition of expectation, in further, we can calculate the expected value of a linear function of random variable.

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The expected value of a linear function $(a + cX)$ of random variable X , constants a and c is

$$E(a+cX) = E(a) + E(cX) = a + cE(X)$$

$$E(aX+cY) = aE(X) + cE(Y)$$

$$E(XY) = E(X)E(Y) \text{ if } X \text{ and } Y \text{ are independent}$$

If $X_1, X_2, \dots, X_i, \dots, X_n$ are n random variables, their

linear function is $\sum_{i=1}^n a_i X_i$, its expected value is

$$E\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i E(X_i)$$

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The variance (方差) of random variable X is defined

$$\sigma^2(X) = E[X - E(X)]^2$$

$$= E(X^2) - [E(X)]^2$$

The variance of linear function $(a + cX)$ of random variable X , constants a and c , is

$$\begin{aligned} \text{Var}(a + cX) &= \text{Var}(a) + \text{Var}(cX) \\ &= c^2 \sigma^2(X) \end{aligned}$$

The variance of a constant is 0.

方差的几种表示: $\sigma^2(X)$, $\text{Var}(X) = \text{Cov}(X, X)$

如不会产生歧义, 有时也用 σ^2 指某一随机变量的方差

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The covariance (协方差) of random variables X and Y is

$$\begin{aligned} \sigma(X, Y) &= E\{(X - E(X))(Y - E(Y))\} \\ &= E(XY - XE(Y) - E(X)Y + E(X)E(Y)) \\ &= E(XY) - E(X)E(Y) \end{aligned}$$

Given constants a, b, c and d , then the covariance of linear function of random variables X and Y is

$$\text{Cov}(a + cX, b + dY) = cd\sigma(X, Y)$$

The covariance of random variables X and X is equal to variance of X

$$\begin{aligned} \sigma(X, X) &= E\{[X - E(X)][X - E(X)]\} \\ &= E(XX) - E(X)E(X) \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

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If $X_1, X_2, \dots, X_i, \dots, X_n$ are n random variables, the variance of their linear function $\sum_{i=1}^n a_i X_i$

$$\begin{aligned} \text{Var}\left(\sum_{i=1}^n a_i X_i\right) &= \text{Cov}\left(\sum_{i=1}^n a_i X_i, \sum_{i=1}^n a_i X_i\right) \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i a_j \text{Cov}(X_i, X_j) \\ &= \sum_{i=1}^n a_i^2 \sigma^2(X_i) + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n a_i a_j \text{Cov}(X_i, X_j) \end{aligned}$$

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If X and Y are independent random variables, then the covariance of X and Y is 0,

$$\text{Cov}(X, Y) = 0$$

Thus, when $X_1, X_2, \dots, X_i, \dots, X_n$ are n independent random variables, then the variance of $\sum_{i=1}^n a_i X_i$

$$\text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \sigma^2(X_i)$$

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The covariance can measure the relationship between different variables, however the value of the covariance will be affected by the unit of measure. The parameter freedom of unit of measure is **correlation coefficient (相关系数)**, it is defined by

$$\rho = \frac{\sigma(X, Y)}{\sqrt{\sigma^2(X)\sigma^2(Y)}} = \frac{\sigma(X, Y)}{\sigma(X)\sigma(Y)}$$

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1.3. Probability Distribution

Probability distribution of discrete variable

Binomial distribution (二项式分布)

In n independent success/failure experiments (Bernoulli trials), if the number of successes of random variable X is x , and the pdf is

$$f(x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} \quad x = 0, 1, 2, \dots, n$$

then random variable X follows binomial distribution. n is the total number of experiments, p ($0 < p < 1$) is the probability of a successful experiment.

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Binomial distribution is often used to describe the distribution of the number of success outcome in n independent experiments.

$X \sim B(n, p)$

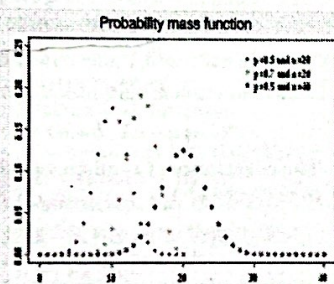
The expected value and variance of random variable X with binomial distribution are

$$E(X) = np = \sum_{i=1}^n i \times f(i)$$

$$\sigma^2(X) = np(1-p)$$

当 $np > 5$ 且 $n(1-p) > 5$ 时, 我们可以使用正态分布来估计二项分布的概率。

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Multinomial distribution (多项分布)

Let x_i indicate the number of i^{th} outcome observed for random variable X in n independent experiments, the joint pdf is

$$f(x_1, x_2, \dots, x_k) = \frac{n!}{\prod_{i=1}^k x_i!} \prod_{i=1}^k p_i^{x_i}$$

Then variable X follows multinomial distribution, n is the total number of experiments, p_i is the probability of i^{th} outcome.

$$X = (X_1 \dots X_i \dots X_k)$$

The expected value and variance of random variable X with multinomial distribution are

$$E(X_i) = np_i \quad \sigma^2(X_i) = np_i(1-p_i)$$

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Poisson distribution (泊松式分布)

If the random variable X could take all non-negative integers, and its pdf is

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

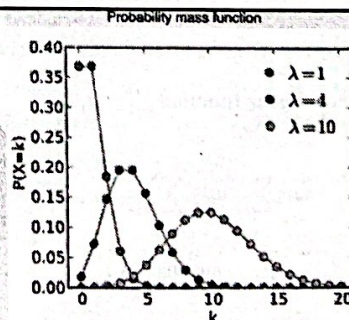
Then variable X follows Poisson distribution, where $e = 2.71828\dots$, parameter λ is the mean times of one event occurring in given time-interval (or area, volume).

Poisson distribution is often used to describe statistical regularity of the occurred times of a specific event in unit time and a range of space.

$X \sim P(\lambda)$

泊松分布: 常用于对不知道事件可能发生总次数的情况下对小概率事件建模, 比如, 交通事故死亡人数, 病毒突变率。

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The expected value and variance of random variable X with Poisson distribution (λ)

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Probability Distribution of Continuous Variable

Normal distribution (正态分布) is an important theoretical distribution of continuous variable, which play an important role in theory and practice of mathematical statistics

If pdf of random variable X is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < +\infty$$

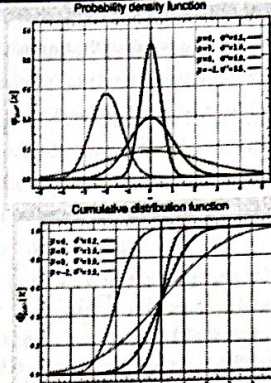
Then variable X follows normal distribution, $e = 2.71828\dots$, parameter μ is mean value, parameter σ^2 is variance.

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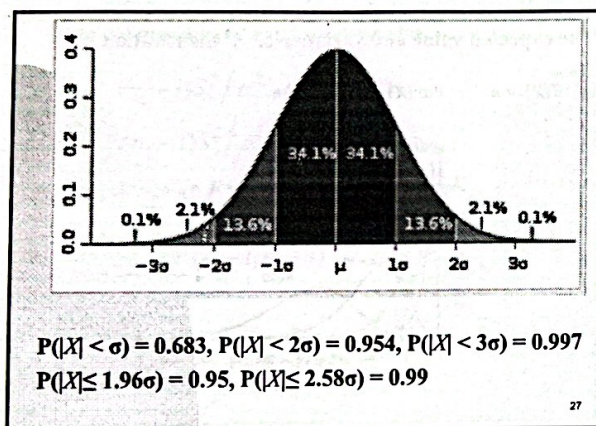
The expected value and variance of random variable X with Normal distribution

$$E(X) = \mu$$

$$\sigma^2(X) = \sigma^2$$

$$X \sim N(\mu, \sigma^2)$$


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Normal Distribution Testing

Skewness (偏度) $skew(X) = E\left[\left(\frac{X-\mu}{\sigma}\right)^3\right]$

- 正态分布的偏度为0，即若数据分布是对称的，偏度为0
- 偏度大于0，则分布偏右，即分布有一条长尾在右
- 偏度小于0，则分布偏左，即分布有一条长尾在左
- 同时偏度的绝对值越大，说明分布的偏移程度越严重

$$\hat{C}_S = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^3}{\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right]^{3/2}}$$

估计偏度的统计量

Kurtosis (峰度) $kurt(X) = E\left[\left(\frac{X-\mu}{\sigma}\right)^4\right]$

- 峰度=0，分布服从正态分布
- 峰度>0，分布陡峭
- 峰度<0，分布平缓

$$\hat{C}_K = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^4}{\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right]^2} - 3$$

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Normal Distribution Testing

If $X \sim N(\mu, \sigma^2)$

Skewness $\hat{C}_S = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^3}{\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right]^{3/2}} \xrightarrow{\text{asymptotically}} N(0, 6/n)$

Kurtosis $\hat{C}_K = \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^4}{\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right]^2} - 3 \xrightarrow{\text{asymptotically}} N(0, 24/n)$

When sample size n is much large, if

$$|\hat{C}_S| > z_{\alpha/2} \sqrt{6/n} \text{ or } |\hat{C}_K| > z_{\alpha/2} \sqrt{24/n}$$

then, we believe that the random variable doesn't follow normal distribution.

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If random variable $X \sim N(\mu, \sigma^2)$, then its linear function $a + cX$ is a new random variable with normal distribution, the mean and variance are $a + c\mu$ and variance $c^2\sigma^2$

$$a + cX \sim N(a + c\mu, c^2\sigma^2)$$

If random variable $\bar{X} \sim N(\mu, \sigma^2/n)$

then the sample mean is also a random variable

$$\bar{X} \sim N(\mu, \sigma^2/n)$$

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Central limit theorem (中心极限定理): for one population following a known distribution with mean μ and variance σ^2 , when sample size n tends to infinite, the sample tends to an asymptotic normal distribution.

A random variable of normal distribution can be transformed into other important probability distribution. Normal random variable $X \sim N(\mu, \sigma^2)$, by following transformation

$$\text{Transformation } z = \frac{X - \mu}{\sigma}$$

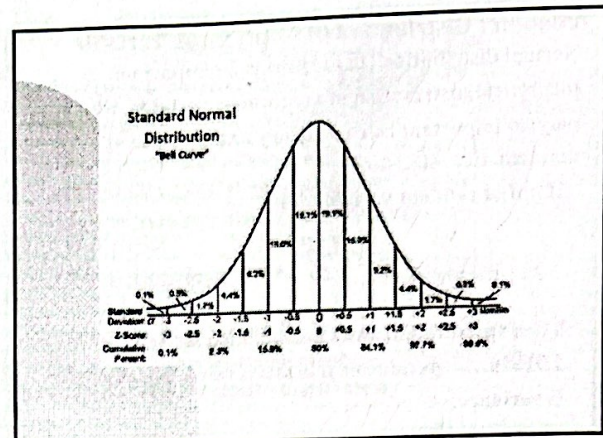
The X is changed to a new random variable z with standard normal distribution $z \sim N(0, 1)$.

$$P(|z| < 1) = 0.6826, P(|z| < 2) = 0.9544, P(|z| < 3) = 0.9974$$

$$P(|z| \leq 1.96) = 0.95, P(|z| \leq 2.58) = 0.9902,$$

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If $z_1, z_2, \dots, z_i, \dots, z_n$ are independent random variables with standard normal distribution, then

$$X = z_1^2 + z_2^2 + \dots + z_n^2$$

satisfy chi-square distribution with degree freedom (df) of n , $X \sim \chi^2(n)$. ν is a unique parameter of the distribution.

If $X \sim \chi^2(n)$, then, its pdf is

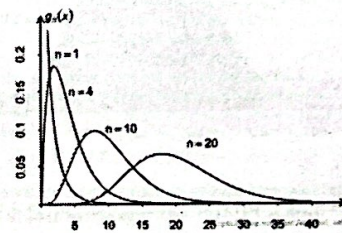
$$f(x) = \begin{cases} \left(\frac{1}{2}\right)^{n/2} \frac{1}{\Gamma(n/2)} x^{n/2-1} e^{-x/2} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

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The expected value and variance of χ^2 distribution

$$E(X) = n \quad \text{Var}(X) = \sigma^2(X) = 2n$$



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If random variable $X \sim N(\mu, \sigma^2)$, its sample standard deviation can be calculated by the formula

$$S = \sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 / (n-1)} \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\text{then, } \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

If random variable $Z \sim N(0, 1)$ and $X \sim \chi^2(\nu)$ are independent, then their function

$$T = \frac{Z}{\sqrt{X/\nu}}$$

follows a t -distribution with df of ν , denoted as $T \sim t(\nu)$

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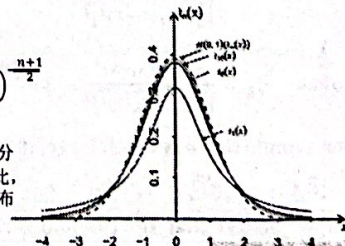
Expected value and variance of T distribution

$$E(T) = 0 \quad \sigma^2(T) = \nu / (\nu - 2)$$

pdf

$$f(x) = \frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi} \Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

当自由度逐渐增大时, t 分布趋近于正态分布, 因此, 大样本时正态分布与 t 分布近似相同



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If random variable $X \sim N(\mu, \sigma^2)$, then, the function of sample mean and deviation follows following distribution

$$\frac{(X - \mu)}{\sigma/\sqrt{n}} \sim z(0, 1) \quad \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

Thus, random variable T

$$T = \frac{\frac{(X - \mu)}{\sigma/\sqrt{n}}}{\sqrt{\frac{(n-1)S^2}{\sigma^2}}} = \frac{(X - \mu)}{S/\sqrt{n}} = t(n-1)$$

satisfy t distribution with df of $(n-1)$

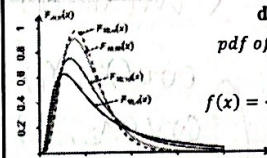
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If random variables $X \sim \chi^2(v_1)$ and $Y \sim \chi^2(v_2)$ are independent chi-square distribution, random variable F as being a function of X and Y

$$F = \frac{X/v_1}{Y/v_2} \sim F(v_1, v_2)$$

satisfy F distribution $F(v_1, v_2)$
 F -distribution has two parameters, namely nominator df v_1 and denominator df v_2 .



pdf of $F(v_1, v_2)$

$$f(x) = \begin{cases} \frac{v_1^{-\frac{v_1}{2}} v_2^{-\frac{v_2}{2}}}{B(\frac{v_1}{2}, \frac{v_2}{2})} x^{\frac{v_1}{2}-1} (1 + \frac{v_1}{v_2} x)^{-\frac{v_1+v_2}{2}} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

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Two independent χ^2 random variables

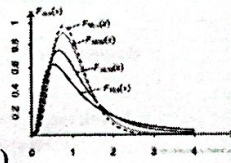
$$(n_1 - 1)S_1^2 / \sigma_1^2 \sim \chi^2(n_1 - 1)$$

$$(n_2 - 1)S_2^2 / \sigma_2^2 \sim \chi^2(n_2 - 1)$$

We can obtain an F variable to test variance

$$F = \frac{S_1^2 / \sigma_1^2}{S_2^2 / \sigma_2^2} \sim F(n_1 - 1, n_2 - 1)$$

$$\text{If } \sigma_1^2 = \sigma_2^2, \quad F = \frac{S_1^2}{S_2^2} \sim F(n_1 - 1, n_2 - 1)$$



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Two random variables satisfying normal distribution, their vector $(X, Y)^T$ is subject to a bivariate normal distribution

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim BVN \left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \sigma_{XY} \\ \sigma_{XY} & \sigma_Y^2 \end{pmatrix} \right) \quad \begin{aligned} \mu_X &= E(X), \mu_Y = E(Y) \\ \sigma_X^2 &= \text{Var}(X), \sigma_Y^2 = \text{Var}(Y), \\ \sigma_{XY} &= \text{Cov}(X, Y) \end{aligned}$$

The correlation coefficient of X and Y

$$\rho = \sigma_{XY} / (\sigma_X \sigma_Y)$$

For each variable, they have marginal distribution, which is normal distribution

$$X \sim N(\mu_X, \sigma_X^2) \quad Y \sim N(\mu_Y, \sigma_Y^2)$$

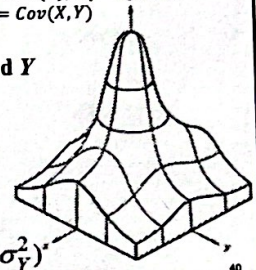


图 1-2 二元正态分布的概率密度函数

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If $Y = y$, the conditional distribution of $X|Y$ is also a random variable with normal distribution

$$X|Y=y \sim N(\mu_X + \rho(\sigma_X / \sigma_Y)(y - \mu_Y), \sigma_X^2(1 - \rho^2))$$

If $X = x$, the conditional distribution of $Y|X$ is also a variable with normal distribution

$$Y|X=x \sim N(\mu_Y + \rho(\sigma_Y / \sigma_X)(x - \mu_X), \sigma_Y^2(1 - \rho^2))$$

边缘分布与条件分布概念上的差别?

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随机向量的期望和方差

$$X = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_m \end{pmatrix}$$

$$Y = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}$$

$$E(X) = \begin{pmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_m) \end{pmatrix}$$

$$\text{Var}(X) = \text{cov}(X, X^T) = \begin{pmatrix} \text{Var}(X_1) & \text{cov}(X_2, X_1) & \vdots & \text{cov}(X_m, X_1) \\ \text{cov}(X_2, X_1) & \text{Var}(X_2) & \vdots & \text{cov}(X_m, X_2) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(X_m, X_1) & \text{cov}(X_m, X_2) & \dots & \text{Var}(X_m) \end{pmatrix}$$

方差-协方差矩阵

$$\text{cov}(X, Y^T) = \begin{pmatrix} \text{cov}(X_1, Y_1) & \text{cov}(X_2, Y_1) & \vdots & \text{cov}(X_m, Y_1) \\ \text{cov}(X_1, Y_2) & \text{cov}(X_2, Y_2) & \vdots & \text{cov}(X_m, Y_2) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(X_1, Y_n) & \text{cov}(X_2, Y_n) & \vdots & \text{cov}(X_m, Y_n) \end{pmatrix}$$

$$\begin{pmatrix} \text{cov}(X_1, X_2) & \dots & \text{cov}(X_1, X_m) \\ \text{Var}(X_2) & \dots & \text{cov}(X_2, X_m) \\ \vdots & \ddots & \vdots \\ \text{cov}(X_m, X_2) & \dots & \text{Var}(X_m) \end{pmatrix}$$

$$\begin{pmatrix} \text{cov}(X_1, Y_2) & \dots & \text{cov}(X_1, Y_n) \\ \text{cov}(X_2, Y_2) & \dots & \text{cov}(X_2, Y_n) \\ \vdots & \ddots & \vdots \\ \text{cov}(X_m, Y_2) & \dots & \text{cov}(X_m, Y_n) \end{pmatrix}$$